

A working review of statistical inference

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The big picture

Today is a fast pass through the whole pipeline of statistical inference — the goal is to see how every piece connects, using **one running example** throughout.

i Our example

A large statistics course. We care about two things: whether a given student **passes** (yes/no), and what **score** a given student earns (a number).

The roadmap:

random variable $\rightarrow$ distribution (discrete & continuous) $\rightarrow$ point estimate $\rightarrow$ sampling distribution $\rightarrow$ Central Limit Theorem $\rightarrow$ hypothesis test $\rightarrow$ t-test

Every step exists to answer one question: **we only ever see one sample — how do we make honest claims about the truth behind it?**

What is a random variable?

Random variable

A numeric quantity whose value depends on the outcome of a random process.

- ▶ We use a **capital letter**, like X , for the random variable itself — the *process* of observing an outcome.
- ▶ We use a **lowercase letter**, like x , for one particular observed value.
- ▶ Two flavors: **discrete** (a countable set of values) and **continuous** (any value in an interval).

In our example: $X = 1$ if a randomly chosen student passes, 0 if not (discrete). $Y =$ a randomly chosen student's exam score (continuous).

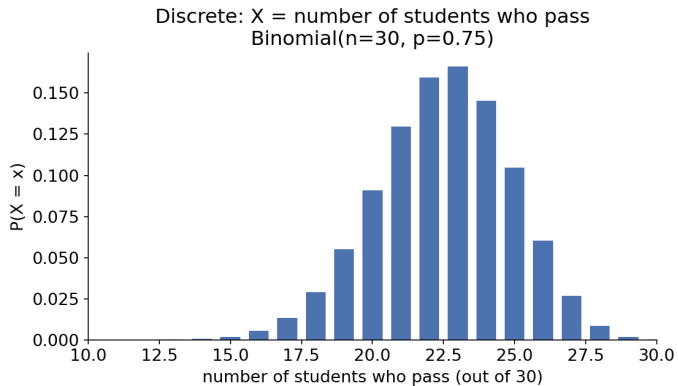
Discrete random variables

For our pass/fail variable X , suppose 75% of students pass ($p = 0.75$). X is a **Bernoulli** random variable.

If we look at $n = 30$ students and count how many pass, that total is **binomial**:

$$P(\text{exactly } k \text{ pass}) = \binom{n}{k} p^k (1-p)^{n-k}, \quad \mu = np, \quad \sigma = \sqrt{np(1-p)}$$

Binomial Visual



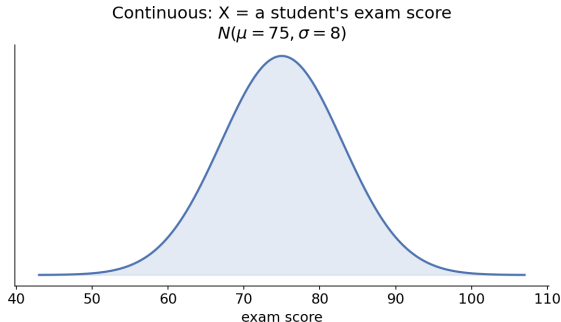
Continuous random variables

Now consider Y = a student's exam score. Y can take *any* value in a range — there's no meaningful “next” value after 74.3.

For a continuous random variable, we describe probability with a **density curve** instead of a bar for each value:

- ▶ The probability of any single exact value is 0.
- ▶ Probabilities come from **areas** under the curve.
- ▶ The most important continuous distribution is the **normal distribution**, $N(\mu, \sigma)$.

The normal distribution



Suppose exam scores follow $N(\mu = 75, \sigma = 8)$.

- ▶ **Standardizing:** $Z = \frac{y - \mu}{\sigma}$ tells you how many standard deviations y is from the mean.
- ▶ The **68-95-99.7 rule**: about 68% of scores fall within 1 SD of 75, 95% within 2 SD, 99.7% within 3 SD.

From individuals to samples: the key pivot

Everything so far described **individual** outcomes. Now the actual problem we care about: we don't know the *true* pass rate or *true* average score for the whole population of students who will ever take this course — we only have **one sample**.

	Population (unknown truth)	Sample (what we observe)
Proportion	p	$\hat{p}$
Mean	μ	$\bar{x}$

$\hat{p}$ and $\bar{x}$ are **point estimates** — our best single guess at an unknown parameter, computed from data.

The sampling distribution

If we gave this exam to a new random sample of 30 students, would $\hat{p}$ come out exactly the same as last time?

No — every new sample gives a slightly different $\hat{p}$ (or $\bar{x}$), just from natural sampling variability. If we imagine repeating this over and over, those estimates form their own distribution: the **sampling distribution**.

- ▶ This isn't just a thought experiment — it's the key idea that lets us quantify *how much* we should trust a single estimate.
- ▶ Its spread has a special name: the **standard error (SE)** — literally the standard deviation of the sampling distribution.

The Central Limit Theorem

i CLT, for a proportion

$$\mu_{\hat{p}} = p, \quad SE_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

i CLT, for a mean

$$\mu_{\bar{x}} = \mu, \quad SE_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

In both cases: when n is large enough and observations are independent, the sampling distribution is approximately **normal**, centered at the truth, with spread that **shrinks as n grows**. This is the single most important fact in this entire lecture — nearly everything else is built on it.

Why does any of this matter?

Here's the actual situation we're always in: we have **one** sample, **one** $\hat{p}$ or $\bar{x}$ — and someone wants to know whether it reflects something *real*, or whether it's just what chance produced.

Suppose 26 of our 30 students pass ($\hat{p} = 0.867$), and historically this course has a 75% pass rate. Is this new group of students doing genuinely better, or is this within the range of ordinary sample-to-sample bounce?

This is a **hypothesis test** — a formal way to make that judgment call, instead of eyeballing it.

Setting up hypotheses

- ▶ H_0 (**null hypothesis**): the skeptical, “nothing unusual is happening” baseline. Here: $p = 0.75$.
- ▶ H_A (**alternative hypothesis**): what we’d conclude if the data are convincing enough. Here: $p \neq 0.75$.

We assume H_0 until the data give us strong reason to abandon it — the same logic as “innocent until proven guilty.” **Failing to reject H_0 never means we’ve proven it true** — only that we don’t have strong enough evidence against it.

One-sided vs. two-sided

- ▶ **One-sided:** H_A claims a specific *direction* ($p > 0.75$, say). Only appropriate when you have a genuine, pre-existing reason to only care about one direction.
- ▶ **Two-sided:** H_A allows *either* direction ($p \neq 0.75$). This is the **default, safer choice** — it keeps you honest about possibilities you didn't expect.

Why it matters

Deciding the direction *after* looking at the data — instead of before — silently inflates your true error rate. Default to two-sided unless you have a specific reason not to.

The test statistic

Whether we're testing a proportion or a mean, the same idea applies: **how many standard errors is our estimate from the null value?**

$$Z = \frac{\text{point estimate} - \text{null value}}{SE}$$

Worked example: testing a proportion

$H_0 : p = 0.75$ vs. $H_A : p \neq 0.75$. Observed $\hat{p} = 26/30 = 0.867$.

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$$SE_{\hat{p}} = \sqrt{\frac{0.75(0.25)}{30}} \approx 0.079$$

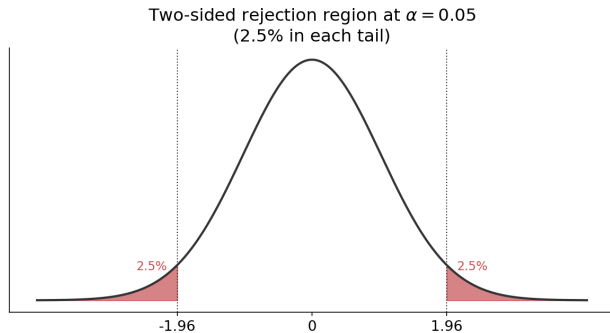
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$$SE_{\hat{p}} = \sqrt{\frac{0.75(0.25)}{30}} \approx 0.079$$

$$Z = \frac{0.867 - 0.75}{0.079} \approx 1.48$$

Worked CI Cont.



Two-sided p-value = $2 \times P(Z > 1.48) \approx 0.139$. Since $0.139 > 0.05$, we **fail to reject** H_0 — not enough evidence that this group's pass rate genuinely differs from 75%.

Confidence intervals: the flip side of a test

A hypothesis test answers yes/no: *is $\hat{p}$ far enough from 0.75 to be convincing?* A **confidence interval** instead asks: *given our data, what's a plausible range for the true p ?*

i Confidence interval, general form

$$\text{point estimate} \pm z^* \times SE$$

where $z^* = 1.96$ for 95% confidence.

Using our pass-rate example ($\hat{p} = 0.867$, $SE_{\hat{p}} \approx 0.079$):

Answer

$$0.867 \pm 1.96(0.079) = (0.712, 1.00)$$

We are 95% confident the true pass rate for this group is between 71.2% and 100%. Since this interval **contains** the null value 0.75, that's consistent with our earlier test's conclusion (fail to reject H_0) — the two methods agree, because they're built from the same SE .

What about testing a mean?

Same logic, new variable: suppose this group of 30 students averaged $\bar{x} = 78$ on the exam, and historically $\mu_0 = 75$.

If we **knew** the population SD ($\sigma = 8$), we could plug straight into the same Z-formula:

$$SE_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{8}{\sqrt{30}} \approx 1.46, \quad Z = \frac{78 - 75}{1.46} \approx 2.05$$

But here's the catch: in practice, we almost never actually know σ . We only have our sample's standard deviation, s — an estimate, with its own uncertainty.

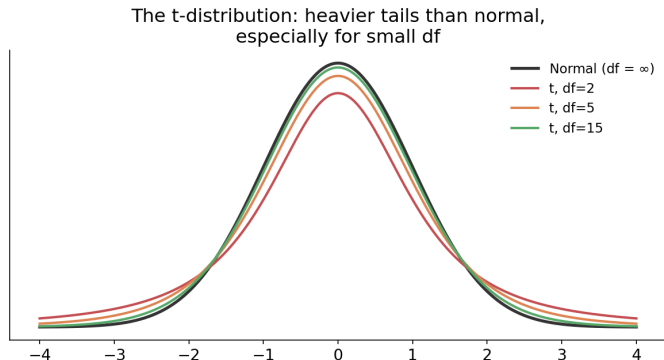
Enter the t-distribution

When we substitute the sample standard deviation s for the unknown σ , we introduce *extra* uncertainty — and the normal distribution is no longer quite the right shape to use.

The t-distribution

Like the standard normal, but with **heavier tails** — reflecting the added uncertainty of estimating σ with s . Indexed by **degrees of freedom**, $df = n - 1$.

T Distribution



As n grows, s becomes a better estimate of σ , and the t-distribution converges to the normal — that's why the two nearly overlap once df is large.

The t-test

i One-sample t-statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad df = n - 1$$

Mechanically **identical** to the Z-formula — same numerator, same idea of “estimate minus null value, divided by standard error” — the only change is *which reference distribution* we compare it to (t instead of normal), because we estimated σ with s .

Worked example: the t-test

Our 30 students: $\bar{x} = 78$, sample SD $s = 8.5$, testing $H_0 : \mu = 75$ vs. $H_A : \mu \neq 75$.

$$SE = \frac{s}{\sqrt{n}} = \frac{8.5}{\sqrt{30}} \approx 1.55, \quad t = \frac{78 - 75}{1.55} \approx 1.93, \quad df = 29$$

Looking up (or using software for) a t-distribution with 29 df: two-sided p-value ≈ 0.063 .

In R: `t.test(scores, mu = 75)` does this entire calculation — statistic, degrees of freedom, p-value, and confidence interval — in one line.

Putting it side by side

	Proportion	Mean
Point estimate	$\hat{p}$	$\bar{x}$
<i>SE</i>	$\sqrt{p(1-p)/n}$	$s/\sqrt{n}$
Reference distribution	Normal	t ($df = n - 1$)
R function	<code>prop.test()</code>	<code>t.test()</code>

Same underlying logic throughout: **(estimate – null value) / SE**, compared against a reference distribution to get a p-value.

Interpreting the result, carefully

- ▶ A p-value is $P(\text{data this extreme} \mid H_0 \text{ true})$ — **not** the probability H_0 is true.
- ▶ “Fail to reject H_0 ” $\neq$ “ H_0 is true” — it means the evidence wasn’t strong enough to abandon our skeptical starting point, given this sample size.
- ▶ Statistical results answer “how surprising is this, assuming nothing unusual is happening?” — they don’t automatically tell you whether an effect is *large* or *practically* important.

Recap: the whole pipeline

1. A **random variable** describes an uncertain numeric outcome — discrete or continuous.
2. A **distribution** (binomial, normal, ...) describes the full range of values it could take and how likely each is.
3. From data, we compute a **point estimate** ($\hat{p}$ or $\bar{x}$) of an unknown population parameter.
4. The **CLT** tells us that estimate's own sampling distribution is approximately normal, for large enough n .
5. A **hypothesis test** asks: is our estimate far enough from a claimed null value to be convincing, given how much natural variability we'd expect?
6. When we must estimate σ with s , we use the **t-distribution** instead of the normal — same logic, adjusted reference distribution.

Every later topic in this course is a variation on this same pipeline.